

# Multi-source multi-terminal reliability evaluation of interconnection networks

Neeraj Kumar Goyal<sup>1</sup> · S. Rajkumar<sup>1</sup>

Received: 11 August 2015 / Accepted: 26 November 2015 / Published online: 16 December 2015  
© Springer-Verlag Berlin Heidelberg 2015

**Abstract** Multistage interconnection networks (MINs) are often used as switching fabrics in parallel and distributed systems designed to provide fast and efficient communication between high-capacity processors. To ensure desired performance of the networks, the reliability evaluation of MIN is quite evident. Reliability studies on MINs evaluated only traditional parameters (two/broadcast/all terminal reliability). With increasing number of input and output nodes in supercomputer environment reliability evaluation of multi-cast nodes is mandatory. A simple and efficient algorithm, Path Tracing Algorithm proposed to trace minimal path sets of any MINs and then associated reliability (or unreliability) expressions are evaluated. The algorithm found quite simple and robust to trace the minimal path sets of a variety of networks and evaluates Multi-source multi-terminal node reliability of any multistage interconnection networks. Total redundant and disjoint paths for each source–destination pair are also evaluated by using the algorithm. Reliability of various MINs are evaluated, analyzed and compared here.

## Abbreviations

|      |                                     |
|------|-------------------------------------|
| MINs | Multistage interconnection networks |
| SEs  | Switching elements                  |
| r(t) | Reliability of switching component  |
| N    | Number of inputs/outputs            |
| S    | Source                              |

|     |                             |
|-----|-----------------------------|
| T   | Terminal                    |
| SVI | Single variable inversion   |
| MVI | Multiple variable inversion |
| LB  | Lower bound                 |
| UB  | Upper bound                 |

## 1 Introduction

Parallel computers with multiple processors have emerged to meet energy-efficient performance demands of current and future challenging computing applications (Wei 1988; Rajkumar and Goyal 2015a, b, c, d; Goyal 2006; Chandrashekar and Goyal 2013; Yang and Wang 2005; Aggarwal and Kaur 2009; Danilewicz and Rajewski 2014; Balkan et al. 2009; Nieminen 2014; Raponi et al. 2013; Pham et al. 2013; He and Delgado-Frias 2007; Fan and Bruck 2000). Supercomputers and multiprocessor systems are emerging communication architecture to address power consumption and performance scaling issues by integrating thousands of processor cores. This cloud computing architecture offer ultrahigh communication bandwidth with low latency, reduced energy consumption and high power efficiency addressing tremendously growing communication demands (Chandrashekar and Goyal 2013; Yang and Wang 2005; Aggarwal and Kaur 2009; Danilewicz and Rajewski 2014; Balkan et al. 2009).

The chip multiprocessors (CMPs) are internally connected via interconnection networks which provide efficient and effective means of communication within many-core system-on-chip (SoC) designs. The processors are interconnected by various designed application-specific network interconnect topologies (Nieminen 2014; Raponi et al. 2013; Pham et al. 2013; Shen et al. 2001; Goyal and Misra 2008; Lee and Hegazy 1998; Chen and Fu 2004; Chen and Chung 2005; Cheng et al. 2006; Chen 2013;

✉ S. Rajkumar  
s.rajkumar@iitkgp.ac.in

Neeraj Kumar Goyal  
ngoyal@hijli.iitkgp.ernet.in

<sup>1</sup> Reliability Engineering Centre, IIT Kharagpur, West Bengal, India

Rajkumar and Goyal 2014). Compatible fabrication process through a networking logic fashion is used to bind these cores. According to the topology, the interconnection networks can be classified into shared medium, direct, indirect and hybrid interconnection networks. These network topologies are designed with predefined application characteristics and traffic patterns.

Thus network-on-chips (NoCs) is a paradigm of network of interconnected nodes to meet growing scalable on-chip communication needs of multi-core processing units (He and Delgado-Frias 2007; Fan and Bruck 2000; Yang and Wang 2004; Shen et al. 2001; Li and Tan 2009). We can interconnect multiple processing nodes, memory controller and input/output devices through bus, crossbar and multistage interconnection networks (MINs). MINs (among indirect networks) are best architectural interconnect trend for multiprocessor systems achieving higher all round performance than bus systems and crossbars.

Multistage interconnection networks (MINs) connect  $N$  number of input nodes to  $N$  number of output nodes through various sizes and various stages of switching elements. The switches are connected to processors/memories through unidirectional/bidirectional links. This scalable alternative MINs are commonly used interconnect architecture in such multi-core processing units connecting processor caches (Jiang et al. 2008; Wei 1988; Rajkumar and Goyal 2015a, b, c, d; Chandrashekar and Goyal 2013; Yang and Wang 2005; Aggarwal and Kaur 2009; Danilewicz and Rajewski 2014; Balkan et al. 2009; Nieminen 2014; Fan and Bruck 2000; Yang and Wang 2004; Shen et al. 2001; Li and Tan 2009; Sengupta and Bansal 2001, 2004; Bansal et al. 1992; Parker and Raghavendra 1984; Lee and Hegazy 1998; Goyal et al. 2005; Rajkumar and Goyal 2015a, b, c, d; Chuang 1996; Chen et al. 2000, 2003; Chen and Chung 2001; Gunawan 2008a).

For effective parallel computing applications MINs will require high performance, yet energy-efficient interconnects. Scaling, fault tolerance and reliability of networks are major performance challenges while addressing these demands. The channel and switch congestion problems escalates when the complexity of the network increases. Thus an efficient MIN should meet various performance parameters like permuting ability, message delay characteristics, throughput, network complexity and cost-effectiveness (Yang and Wang 2005; Aggarwal and Kaur 2009; Danilewicz and Rajewski 2014; Balkan et al. 2009; Nieminen 2014; Raponi et al. 2013; Pham et al. 2013; He and Delgado-Frias 2007; Fan and Bruck 2000).

The switch/link failures lead to dilapidation of network interconnections which ends in unbalanced traffic load. To achieve this they require high fault-tolerance, yet reliability-efficient interconnects. A reliable interconnect is a promising solution in building a fault-tolerant system to

provide successful transmission (Wei 1988; Rajkumar and Goyal 2014, 2015a, b, c, d; Goyal 2006; Rajkumar 2015a, b; Gunawan 2008b; Blake and Trivedi 1989; Koren and San Francisco 2010; Shooman 2001; Gunawan 2013; Verlinden et al. 2012; Bobbio et al. 2011). A network designer has to choose a suitable fault tolerant and highly reliable MIN architecture to avoid performance degradation and to route multiple communication tasks concurrently.

- Reliability and fault tolerance of MINs has long been a concern of researchers.
- According to the reported researches (Rajkumar and Goyal 2014, 2015c; Gunawan 2008a, b; Gunawan and Fard 2012; Blake and Trivedi 1989; Koren and San Francisco 2010; Shooman 2001) Wang and Mingxiao 2004; Distefano 2009) and literatures there is very limited reliability comparison of MIN architectures.
- Some literatures calculated two-terminal network reliability (TTNR) commonly known as the source-terminal reliability.
- Other researchers extend to  $k$ -terminal network reliability (KTNR) and all-terminal network reliability (ATNR).
- However we have extended the traditional reliability evaluation to include a new reliability measure as multi-source multi-terminal reliability (MSMT).

This paper is organized as follows: Sect. 2 surveys definitions and motivations. Section 3 presents the overview of the algorithm and Sect. 4 describes the reliability evaluation. Concluding remarks are provided in Sect. 5, followed by references.

## 2 Background

### 2.1 Basic definitions

The reliability evaluation of MIN can be sub-divided into following different cases (Goyal and Misra 2008; Goyal et al. 2005; Rajkumar and Goyal 2014; Gunawan 2008a, b; Gunawan and Fard 2012; Birolini 2007; Gunawan 2013; Verlinden et al. 2012; Bobbio et al. 2011):

- Two terminal or terminal pair reliability (TPR): To send messages from a source node  $S$  to a terminal node  $T$ . The terminal pair reliability of a network is defined as the probability of having at least one operational path between the nodes  $s$  and  $t$ .
- Global or all terminal reliability (ATR): Another communication operation is complete network connectivity. The all terminal reliability of a network is defined as the probability that for every node pair there exist an

operational path to connect them; or equivalently, the probability that there exist a working spanning tree.

- Broadcast reliability: The probability that a single source node is able to broadcast data to all terminal nodes.
- Multi-source-multi terminal reliability (MSMTR): The probability of successful communication between given set of multiple source nodes (S) and multiple terminal nodes (T) in the network. Redundant paths (Rajkumar and Goyal 2014, 2015a, b, c) are all possible paths for any source destination nodes pair.
- Disjoint paths are redundant paths which do not have any element in common other than source and destination nodes.

## 2.2 Motivation

Available literature (Rajkumar and Goyal 2015c) on reliability evaluation of MINs use various techniques like

- Continuous time Markov chains (CTMC) (Blake and Trivedi 1989)
- Reliability bounds (Wei 1988; Gunawan 2008a)
- Reliability block diagram (Gunawan 2008b; Gunawan and Fard 2012; Distefano 2009)
- Decomposition method (Koren and San Francisco 2010; Shoorman 2001; Wang and Mingxiao 2004)
- Monte Carlo simulation (Gunawan 2013)

*However there is always a demand in finding the exact reliability of multistage interconnection networks for both small and large scale complex parallel interconnection systems.*

In order to accurately determine reliability of MINs from its components, a simple, general and efficient algorithm for exact reliability of MINs is needed.

Detailed review of MINs on fault tolerance, reliability evaluations and their drawbacks is provided in (Rajkumar and Goyal 2015a, c).

## 2.3 Continuous time Markov chains (CTMC)

It is applied for simple networks. This approach provides exact reliability. In Markov modeling a set of finite number of states called a Markov chain are used to depict the behavior of the system.

The size of the state space of a Markov chain grows exponentially with the size of the network. For smaller network size, CTMC reliability modeling of MIN (Blake and Trivedi 1989) is very simple but they are complex when network size increases due to exponential growth of the state space.

For example: we have to consider  $2^{16}$  possible states for  $8 \times 8$  Omega + network designed using 16 switching elements (SE).

## 2.4 Reliability bounds

They are also termed as lower and upper bounds (LUB) method.

Reliability bound method (lower and upper bounds) is a better alternative to estimate MIN reliability with larger number of inputs. Reliability block diagram can be used to aid the derivation of bound equations of network reliability.

Upper bound value represents the optimistic view of the network for successful interconnection. Finding an upper bound for system reliability is usually not the important reliability criteria because we usually want a conservative indication of how long the system will be operational. This pictures the importance of lower bound estimation.

Lower bound value represents the pessimistic view and provides sufficient and reasonable approximation to the actual network reliability such that the system will be operational over the specified time interval. But when complexity increases, these methods fail to attain accurate results in general.

## 2.5 Monte Carlo simulation

Monte Carlo methods (Gunawan 2013) have been used to evaluate MINs reliability. These methods perform simple random sampling of switching element (SE) into two states as working or failure states. These states are then considered to determine network state as failure or success. Mean of numerous runs is taken to get approximate reliability value considering ratio of number of success to total samples. Accuracy of reliability estimation can be increased by increasing the number of samples.

However, increasing the number of replications will indirectly increase computational resource and time requirements substantially. The sample size requirement is further very high for high reliability switches as they require large number of samples to obtain appropriate representation of switch failures.

Estimation of reliability using Monte Carlo method becomes time consuming and rather complex when the network size increases. They provide reliability estimation based on sampling randomization and do not produce accurate reliability values.

Monte Carlo simulation use is generally limited to small networks like Shuffle exchange networks (Gunawan 2013). Therefore MIN reliability estimation of larger networks usually utilizes approximation techniques to determine the overall system reliability. When complexity of MINs increases, the accuracy of evaluated reliability values reduces.

## 2.6 Reliability block diagram

Reliability block diagram (RBD) is another approach used to estimate network reliability of MIN (Gunawan 2008b;

Gunawan and Fard 2012; Distefano 2009; Birolini 2007). Reliability block diagram can be used to aid the derivation of bound equations which is an approximation technique for determining MIN reliability.

Individual works presented in literatures in general, calculated one of the reliability parameters (TPR/ATR/KTR) for the common MIN like Omega network (Shen et al. 2001), Shuffle exchange network (Kumar and Reddy 1987) and Cube network (Fan and Bruck 2000).

- All terminal reliability parameters need to be evaluated for complicated switching topology like gamma network (Parker and Raghavendra 1984), balanced GIN (Cheng et al. 2006) and incomplete GIN (Chen 2013).
- Reliability of major recently developed MINs (Danilewicz and Rajewski 2014; Pham et al. 2013; Chiuyuan and Jing-Kai 2006; Cheng et al. 2006; Chen 2013) are yet to be evaluated.
- Redundant and disjoint paths need to be evaluated for all MINs.
- Reliability evaluation methods should compute exact reliability value

Reliability bounds and RBD: Network size increases and for complex networks they cannot attain accurate results.

Monte Carlo method: Several runs are used to have a better approximation value.

Markov chain approach: Network size increases, they are arduous and complex.

There is lack of papers investigating all the reliability parameters for all MINs. It clearly motivates us to study MINs for achieving a comprehensive reliability analysis and comparison.

Thus there is a need of a single algorithm to accurately evaluate all reliability measures of MINs. This motivated us to implement a simple, versatile and efficient algorithm for exact reliability evaluation of MINs.

## 2.7 Contribution

The frequently considered network reliability measures are: 2-terminal, K-terminal and all-terminal reliability. Recently (Bobbio et al. 2011) calculated the reliability analysis of multi source and multi terminal critical interacting systems using BDD paved the way to evaluate multi-source multi-terminal MIN reliability. In context of MINs, multi-source multi-terminal reliability is defined as the probability that a combination of terminal nodes is connected to a combination of source nodes for given period of time under given use conditions.

Previous works has been limited only to computing more general and traditional measures of reliability. Determination of and disjoint path information of MINs.

This is the very first attempt to find redundant paths and disjoint paths of MINs and calculate reliability measures of MINs. Basic idea for the fault-tolerance of the MIN is to create redundant (alternative) paths between any source–destination (S-D) node pairs, so that redundant paths can be used when any switch gets failed/busy. Fault-tolerance is analyzed in terms of number of redundant and disjoint paths.

Providing more disjoint paths enhance MIN fault tolerance as the common elements leading to multiple redundant path failures are absent. Therefore, it gives better information about network capability to withstand minimum number of switch failures. Both redundant and disjoint paths are calculated using proposed MSMT algorithm and presented in Table 7 for various MINs.

This is the list of options available in proposed algorithm to evaluate MIN reliability:

- Two-terminal reliability
- K-terminal reliability
- All-terminal reliability
- Broadcast reliability
- Multi-source multi-terminal reliability
- Numeric result regarding both Redundant and disjoint paths of MINs.

## 3 Multi source-multi terminal reliability (MSMT) algorithm

A major problem with accurate computation of reliability measures for a large and complex network, is the required computation time and complexity of analysis (Wei 1988; Rajkumar and Goyal 2014, 2015a, b, c, d; Gunawan 2008b; Gunawan 2013; Verlinden et al. 2012; Bobbio et al. 2011).

It is difficult to compute the reliability of real life communication networks, which are quite large, on desktop computers due to long execution time and high memory requirements. Such computations are usually performed on high-end processors for critical systems. An extensive research has been carried out to minimize the efforts required for these computations and develop useful tools for network designers.

In recent years parallel and distributed computing dependent on computer communication networks. Moreover, it is a well-known fact that every component or system is prone to failure and probability of system failure increases with increase in the system size and complexity. Therefore it is very essential to compute and ensure the reliability of these networks, growing complex day by day.

Reliability modeling and computation is necessary for analyzing these networks (Goyal 2006; Rajkumar and Goyal 2014, 2015a, b; Blake and Trivedi 1989; Koren and

San Francisco 2010; Shooman 2001; Gunawan 2013; Verlinden et al. 2012; Bobbio et al. 2011) and improve the weak links using reliability and design techniques. Considering the present developments in communication hardware industry, where the network components are becoming more reliable and non-repairable, reliability becomes an important performance measure for computer communication networks.

A multistage interconnection network is modeled by a probabilistic connected graph (Gunawan 2008b; Gunawan and Fard 2012; Blake and Trivedi 1989; Koren and San Francisco 2010; Shooman 2001; Wang and Mingxiao 2004). Multistage interconnection networks (MINs) connect  $N$  number of input nodes to  $N$  number of output nodes through various sizes and various stages of intermediate switching elements. MINs are stage wise networks. MIN has structure in which connections exist to same layer or adjacent layer only. Connections between non-adjacent layers do not exist.

MSMT algorithms enumerate all minimal paths for 2-TR, broadcast and ATR and also evaluate the MINs reliability with multiple source–destination (S-D) nodes.

Reliability evaluation of interconnection networks can broadly be classified into two paradigms:

### 3.1 Path sets or cut sets approaches (POC) paradigm (Goyal 2006)

These use path sets or cut sets as the starting point for TPR problems, spanning trees for ATR problems and  $k$ -trees for KTR problem.

A minimal path set (Gunawan 2008a, b) is generally used in the sense of a minimal chain of elements required to establish continuity between the two specified terminals (nodes or vertices).

The network reliability can also be computed indirectly by taking advantages of duality in graphs. In other words, instead of computing reliability one can compute unreliability in the sense of a minimal set of elements whose failure ensures network failure. This set is known as minimal cut set of the network. Therefore, success of all the components in at least one minimal path set ensures network success whereas failure of all components in at least one minimal cut set ensures network failure.

The POC approaches determine reliability (unreliability) expression in the disjoint set form. These approaches can be classified as: (1) approaches which give disjoint terms in SVI (single variable inversion) form and (2) approaches which give disjoint terms in MVI (multiple variable inversion).

The MVI approaches have advantage over SVI of giving more compact reliability expression. However, SVI

approaches use a simple data presentation that makes these approaches more practical for storing and processing the data.

Several methods have been proposed in literature (Goyal 2006; Chandrashekar and Goyal 2013; Goyal and Misra 2008; Goyal et al. 2005; Gunawan 2008a) to enumerate path sets (cut sets) for TPR evaluation of a given network. Goyal (2006) and Gunawan (2008) has described an approach to determine min-tie-sets of complex networks.

### 3.2 Non-path sets or cut sets approaches (NPOC) paradigm (Goyal 2006)

These approaches do not use path sets or cut sets rather these apply reduction/decomposition or transformation or a combination of these approaches. The non-path (or cut) set approaches may be categorized as state enumeration methods, topological methods, transformation methods and decomposition (factoring theorem) methods.

The NPOC approaches, generally, operate on component reliability values directly and do not provide a reliability expression. Therefore, change in reliability value of even a single component requires repetition of the whole procedure to calculate a new reliability value of the network. Moreover, a designer cannot efficiently analyze the dependency of network reliability on its components.

Path set and cut set techniques are the most widely used techniques for terminal pair reliability evaluation of Interconnection networks as these techniques give a compact reliability expression in sum of disjoint product (SDP) form. These techniques first enumerate minimal path (or cut) set for the network and then evaluate reliability (or unreliability) expressions in the SDP form. Since enumeration of cut set also requires knowledge of basic paths of the network, the path set enumeration becomes a necessity for these techniques.

MSMT MIN reliability is evaluated in three steps:

- Path tracing algorithm (PTA) which enumerates path set of a system in a described format
- Path retrieval algorithm (PRA) retrieves paths from the formatted output of PTA
- Reliability expressions from these paths set are evaluated by using hybrid multiple variable inversion (MVI) algorithm (Chaturvedi and Misra 2002)

Two stepwise algorithms namely the PTA and PRA are presented in this section. Application of these algorithms, in sequence, gives path set of a network. The PTA and PRA are described by flow charts shown in Figs. 1 and 2 respectively.



### 3.3 Path tracing algorithm (PTA)

The variables to recursive function PTA are initialized as follows:

1. Make the **Previous node** equal to zero as branching has not started yet.
2. Initialize **Counter** to 1.
3. Initialize **NP** to 0.
4. Make the source node row as **Current row**.

**Inputs** are: CM (connection matrix), Previous node, Counter, and Current row.

**Global variables** are: Component, Branch number, Counter, Parent branch, End Branch, and NP (number of paths)

1. Set **Start counter** equal to **Counter** value.
2. Read the columns of **Current row**, one by one, as **new element**.
3. Leave out the **diagonal element**.
4. If **new element** is zero, ignore it.
5. Store the other elements.
6. Increment the counter once for each element stored.
7. **Branch number** of the element read will be the current value of the **Counter**.
8. The **Parent branch** of each new element will be equal to the number stored in the **previous node**.
9. The **Node number** of each element is the column number of the element read.

Repeating above listed sequence (step 2 to step 9), all necessary components are read.

10. Modify connection matrix **CM** by changing all entries in column, corresponding to **Current row**, to zero.
  11. Set **End counter** equal to **Counter - 1**.
  12. Initialize  $j = \text{Start counter}$ .
  13. If  $j$  is greater than **End counter**, then RETURN from the current recursive procedure **PTA**.  
Otherwise, read **Node number** ( $j$ ).
  14. If **Node number** ( $j$ ) is output node then store value of  $j$  in **End Branch**, increase **NP** and  $j$  by 1, and go to step 13.
- Otherwise, make a new recursive call to procedure **PTA** assigning new input variables as:
- **Previous node** = **Branch number** of the element.
  - **Current row** = **Node number** of the element.
  - **CM** = **CM**

Then increment  $j$  by 1 and repeat step 13.

When all the calls from recursive procedure are returned, it means that all minimal paths have been traced. Then go to PRA for path retrieval from the output of PTA.

### 3.4 Path retrieval algorithm (PRA)

The input variables to PRA are the global variables resulted from PTA. These are Component, Branch number, Parent branch, End Branch, and NP (number of paths).

1. Set index variable  $i = 1$ .
2. If  $i > \text{NP}$ , then **STOP**. All paths are retrieved.
3. Read component **End Branch** ( $i$ ) from stored components.
4. If **Component** is equal to 1, then do not store component name. It only indicates a short.
5. Else, store the element.
6. If **Branch Number**, corresponding to the element, is equal to 0, then increase  $i$  by 1 and go to step 2.
7. Else, read element for branch number equal to **Parent Branch** number and go to step 4.

All paths are retrieved from the stored format.

## 4 Results and discussion

Algorithms PTA and PRA are applied to determine min-*t*-sets for node pair (1, 5) (**Two Terminal**) of the sample

network shown in Fig. 3. The connection matrix (CM) presentation is shown in Table 1.

Execution results of PTA are shown in Table 2 in a formatted sequence of variable arrays. The intermediate connection matrices used in PTA are displayed in Table 3. An execution sequence for recursive calls and returned calls of procedure PTA is given in Table 4. In this table, **Ret** corresponds to call returned from PTA and **New** corresponds to new recursive call made to PTA. Working of PTA is explained by referring Tables 2, 3, 4.

Initially input variables to PTA are (refer to row#1 in Table 4):

- **CM** as shown in Table 1; **Previous Node** = 0; **Current Row** = 1.

Execution of PTA, up to 10 steps, adds branches from #1 to #3 shown in Table 2. These are the elements present in **Current Row** (row#1) of **CM**. Then step 11 modifies the **CM** as shown in Table 3a by making all elements present in column corresponding to **Current Row** equal to 0. Then execution of step 12 and 13 set **Counter** for execution of step 14 and 15 for all added branches (#1 to #3). First added branch is #1 and it doesn't terminate to **sink node** therefore a recursive call to PTA is made with inputs as below (refer to row#2 in Table 4)

- **CM** as shown in Table 3a; **Previous Node** = 1; **Current Row** = 2.

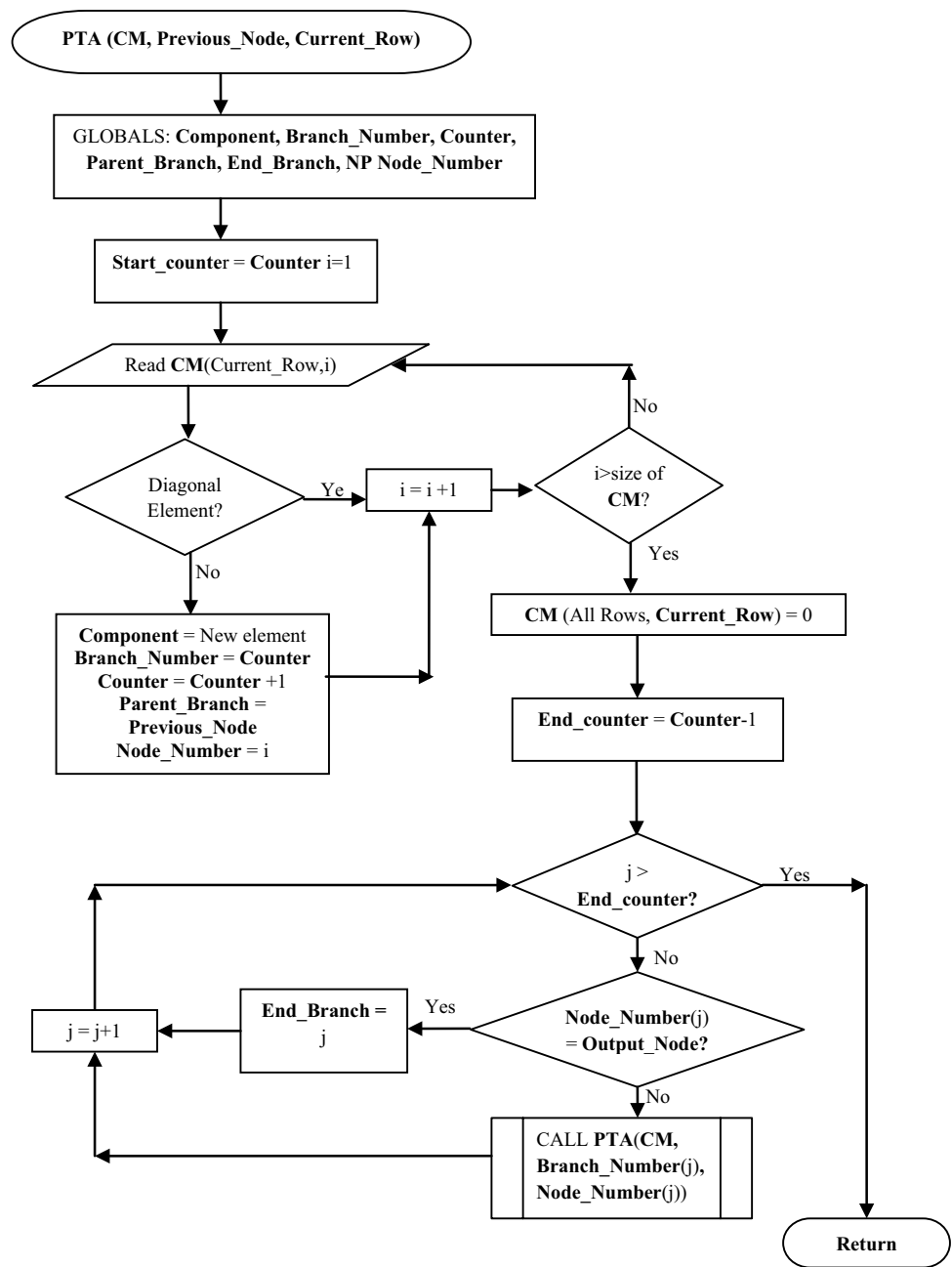
First 10 steps execution of PTA result in adding components present in row#2 of Table 3a. It results in addition of branches from #4 to #6 and modification of **CM** as shown in Table 3b by changing all elements present in column#2 (corresponding to **Current Row**) of Table 3a to zero. Added branch #4 doesn't terminate to **sink node** therefore another recursive call to PTA is made with inputs as below (refer to row#3 in Table 4)

- **CM** as shown in Table 3b; **Previous Node** = 4; **Current Row** = 3.

This recursive call results in addition of two more branches #7 and #8 and **CM** is modified as shown in Table 3c by changing elements present in column#3–0. Added branch #7 leads to another call of PTA with inputs as below (refer to row#4 in Table 4)

- **CM** as shown in Table 3c; **Previous Node** = 7; **Current Row** = 4.

This call doesn't result in any addition of branches since no element is present in row 4 of Table 3c. Therefore the call is returned and **CM** of Table 3c is restored. Then PTA,

**Fig. 1** Flowchart of the proposed recursive procedure PTA

corresponding to returned call, continues for branch #8. Since branch #8 terminates to sink node, its number is added to *End Branch*, which stores the termination points of the branches. There are no other branches added in this call of the procedure therefore it returns. It results in restoration of *CM* shown in Table 3b. It makes another recursive call corresponding to next branch #5 with inputs as below (refer to row#5 in Table 4):

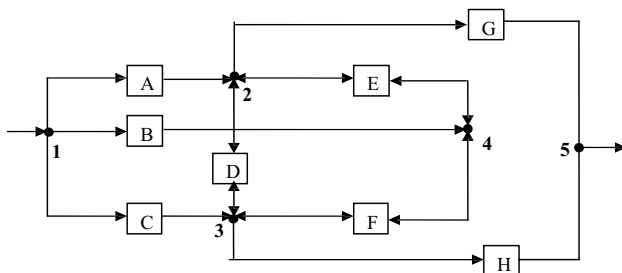
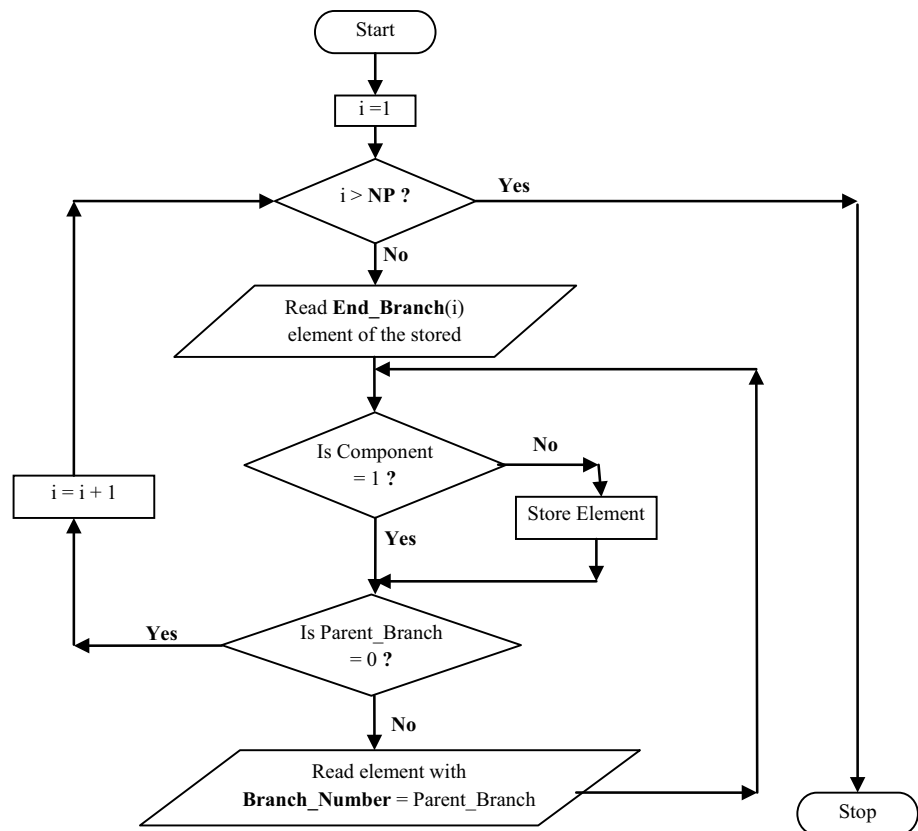
**CM** as shown in Table 3b; **Previous Node** = 5; **Current Row** = 4.

Row#4 of Table 3.4(b) has one element therefore one branch #9 is added. This branch makes another recursive call with inputs as below (refer to row#6 in Table 4):

- **CM** as shown in Table 3d; **Previous Node** = 9; **Current Row** = 3.

Row#3 of Table 3d adds another branch #10 terminating to sink node. Therefore branch #10 is added to *End Branch*. Two recursive calls are then returned and *CM* shown in Table 3b is restored. Last added branch #6 in that call terminates to *sink node*. Therefore #6 is added to *End Branch*, another call is returned and *CM* shown in Table 3a is restored to find leading branches from branch #2 and branch #3.

Similar to above, rest of the branches, shown in Table 2 are determined. After complete execution of

**Fig. 2** Flowchart for path retrieval algorithm (PRA)**Fig. 3** Sample network**Table 1** Connection matrix (CM)

| Nodes | 1 | 2 | 3 | 4 | 5 |
|-------|---|---|---|---|---|
| 1     | 1 | A | C | B | 0 |
| 2     | 0 | 1 | D | E | G |
| 3     | 0 | D | 1 | F | H |
| 4     | 0 | E | F | 1 | 0 |
| 5     | 0 | 0 | 0 | 0 | 1 |

PTA, PRA is called to retrieve paths from the stored formatted data.

**Path Retrieval Algorithm (PRA)** PRA reads min-tie-set of the system from the formatted data stored in Table 2.

Number of paths in the tie-set is equal to  $NP$ , which is equal to the number of entries in array *End Branch*. Each entry in *End Branch* is taken as starting point and then it trace back through variable *Previous Node* to find components present in corresponding path.

Table 5 shows the tracing sequence of branches for all retrieved paths. It can be observed, from Table 5, that all paths have starting branch corresponding to the entries of array *End Branch*, shown in the last column of Table 4.

Thus 10 path sets are enumerated for the sample network shown in Fig. 3. Reliability expressions are then evaluated by using hybrid multiple variable inversion (MVI) algorithm (Chaturvedi and Misra 2002).

Following same fashion we evaluated the reliability of various MIN topologies in the forthcoming section and shown in Tables 7, 8, 9.

## 5 Reliability analysis and comparison of MINs

In this section all reliability parameters of various MINs are evaluated using MSMT algorithm and compared. Two-terminal network reliability (TTNR) values are shown in Table 7. All-terminal network reliability (ATNR) and broadcast reliability values are shown in Table 7.



**Table 2** Complete tie-set in formatted form after application of PTA

| Branch Number | 1  | 2         | 3  | 4         | 5  | 6        | 7  | 8         | 9         | 10       | 11        | 12        | 13       |
|---------------|----|-----------|----|-----------|----|----------|----|-----------|-----------|----------|-----------|-----------|----------|
| Parent Branch | 0  | 0         | 0  | 1         | 1  | <b>1</b> | 4  | <b>4</b>  | 5         | <b>9</b> | 2         | 2         | <b>2</b> |
| Component     | A  | C         | B  | D         | E  | <b>G</b> | F  | <b>H</b>  | F         | <b>H</b> | D         | F         | <b>H</b> |
| Node Number   | 2  | 3         | 4  | 3         | 4  | <b>5</b> | 4  | <b>5</b>  | 3         | <b>5</b> | 2         | 4         | <b>5</b> |
| Branch Number | 14 | 15        | 16 | 17        | 18 | 19       | 20 | 21        | 22        | 23       | 24        | 25        |          |
| Parent Branch | 11 | <b>11</b> | 12 | <b>16</b> | 3  | 3        | 18 | <b>18</b> | <b>20</b> | 19       | <b>19</b> | <b>23</b> |          |
| Component     | E  | <b>G</b>  | E  | <b>G</b>  | E  | F        | D  | <b>G</b>  | <b>H</b>  | D        | <b>H</b>  | <b>G</b>  |          |
| Node Number   | 4  | <b>5</b>  | 2  | <b>5</b>  | 2  | 3        | 3  | <b>5</b>  | <b>5</b>  | 2        | <b>5</b>  | <b>5</b>  |          |

Bold values indicate the PTA results

**Table 3** Intermediate matrices in PTA

| (a) |   |   |   |   |   | (b) |   |   |   |   |   | (c) |   |   |   |   |   |
|-----|---|---|---|---|---|-----|---|---|---|---|---|-----|---|---|---|---|---|
| #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 |
| 1   | 0 | A | C | B | 0 | 1   | 0 | 0 | C | B | 0 | 1   | 0 | 0 | 0 | B | 0 |
| 2   | 0 | 1 | D | E | G | 2   | 0 | 0 | D | E | G | 2   | 0 | 0 | 0 | E | G |
| 3   | 0 | D | 1 | F | H | 3   | 0 | 0 | 1 | F | H | 3   | 0 | 0 | 0 | F | H |
| 4   | 0 | E | F | 1 | 0 | 4   | 0 | 0 | F | 1 | 0 | 4   | 0 | 0 | 0 | 1 | 0 |
| 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 |
| (d) |   |   |   |   |   | (e) |   |   |   |   |   | (f) |   |   |   |   |   |
| #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 |
| 1   | 0 | 0 | C | 0 | 0 | 1   | 0 | A | 0 | B | 0 | 1   | 0 | 0 | 0 | B | 0 |
| 2   | 0 | 0 | D | 0 | G | 2   | 0 | 1 | 0 | E | G | 2   | 0 | 0 | 0 | E | G |
| 3   | 0 | 0 | 1 | 0 | H | 3   | 0 | D | 0 | F | H | 3   | 0 | 0 | 0 | F | H |
| 4   | 0 | 0 | F | 0 | 0 | 4   | 0 | E | 0 | 1 | 0 | 4   | 0 | 0 | 0 | 1 | 0 |
| 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 |
| (g) |   |   |   |   |   | (h) |   |   |   |   |   | (i) |   |   |   |   |   |
| #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 |
| 1   | 0 | A | 0 | B | 0 | 1   | 0 | A | 0 | 0 | 0 | 1   | 0 | A | C | 0 | 0 |
| 2   | 0 | 1 | 0 | E | G | 2   | 0 | 1 | 0 | 0 | G | 2   | 0 | 1 | D | 0 | G |
| 3   | 0 | D | 0 | F | H | 3   | 0 | D | 0 | 0 | H | 3   | 0 | D | 1 | 0 | H |
| 4   | 0 | E | 0 | 1 | 0 | 4   | 0 | E | 0 | 0 | 0 | 4   | 0 | E | F | 0 | 0 |
| 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 |
| (j) |   |   |   |   |   | (k) |   |   |   |   |   | (l) |   |   |   |   |   |
| #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 | #   | 1 | 2 | 3 | 4 | 5 |
| 1   | 0 | 0 | C | 0 | 0 | 1   | 0 | A | 0 | 0 | 0 | 1   | 0 | 0 | 0 | 0 | 0 |
| 2   | 0 | 0 | D | 0 | G | 2   | 0 | 1 | 0 | 0 | G | 2   | 0 | 0 | 0 | 0 | G |
| 3   | 0 | 0 | 1 | 0 | H | 3   | 0 | D | 0 | 0 | H | 3   | 0 | 0 | 0 | 0 | H |
| 4   | 0 | 0 | F | 0 | 0 | 4   | 0 | E | 0 | 0 | 0 | 4   | 0 | 0 | 0 | 0 | 0 |
| 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 | 5   | 0 | 0 | 0 | 0 | 1 |

The reliability values are calculated by assuming different switch reliability values ( $r = 0.99$  and  $0.90$ ). Finally multi-source multi-terminal reliability (MSMT) evaluations are shown in Tables 8, 9 for node reliability value ( $r = 0.95$  and  $0.50$ ). Following general assumptions are made for the terminal reliability evaluation in this work (Goyal and Misra 2008; Goyal et al. 2005; Rajkumar and Goyal 2014; Gunawan 2008a, b; Gunawan and Fard 2012)

- All failures are statistically independent
- The links of the network are perfectly reliable
- The switch has only two states (1) working or (2) failed. Partial functionality loss of switches is considered as failure of switch.

## 5.1 Network representation

4-Disjoint paths gamma interconnection network named (4DGIN-1) proposed in (Rajkumar and Goyal 2014) and shown in Fig. 2. 4DGINs is a multipath GIN providing four disjoint paths. 4DGINs has full access property, i.e., it provides connection from any source nodes to any destination nodes.

We consider 4-disjoint paths gamma interconnection network (4DGIN-1) is generally represented having **8 Sources and 8 Destinations** connected by **24 switches** and **72 links** are stored in **Connection Matrix or CM**.

4DGIN-1 network is represented or fed to a computer is its connection matrix. Connection matrix is ' $N \times N$ ' matrix, whose N rows corresponds to N nodes (switches

**Table 4** Complete execution flow of PTA

| #  | Recursive call     | Inputs to recursive call |               |             | Added branches | Added end branches |
|----|--------------------|--------------------------|---------------|-------------|----------------|--------------------|
|    |                    | CM (Table)               | Parent branch | Current row |                |                    |
| 1  | New                | Table 1                  | 0             | 1           | {1,2,3}        | –                  |
| 2  | New                | Table 3a                 | 1             | 2           | {4,5,6}        | –                  |
| 3  | New                | Table 3b                 | 4             | 3           | {7,8}          | –                  |
| 4  | New                | Table 3c                 | 7             | 4           | –              | –                  |
| 5  | Ret, Ret, New      | Table 3b                 | 5             | 4           | {9}            | {8}                |
| 6  | New                | Table 3d                 | 9             | 3           | {10}           | –                  |
| 7  | Ret, Ret, Ret, New | Table 3a                 | 2             | 3           | {11,12,13}     | {10,6}             |
| 8  | New                | Table 3e                 | 11            | 2           | {14,15}        | –                  |
| 9  | New                | Table 3f                 | 14            | 4           | –              | –                  |
| 10 | Ret, Ret, New      | Table 3g                 | 12            | 4           | {16}           | {15}               |
| 11 | New                | Table 3h                 | 16            | 2           | {17}           | –                  |
| 12 | Ret, Ret, Ret, New | Table 3a                 | 3             | 4           | {18,19}        | {17,13}            |
| 13 | New                | Table 3i                 | 18            | 2           | {20,21}        | –                  |
| 14 | New                | Table 3j                 | 20            | 3           | {22}           | –                  |
| 15 | Ret, Ret, Ret, New | Table 3k                 | 19            | 3           | {23,24}        | {22,21}            |
| 16 | New                | Table 3l                 | 23            | 2           | {25}           |                    |
| 17 | Ret, Ret, Ret      |                          |               |             |                | {25,24}            |

**Table 5** Paths obtained from Table 2 after application of PRA

| Path # | Components in path | Sequence of branch numbers followed |
|--------|--------------------|-------------------------------------|
| 1      | HDA                | 8 -> 4-> 1                          |
| 2      | HFEA               | 10 -> 9-> 5-> 1                     |
| 3      | GA                 | 6 -> 1                              |
| 4      | GDC                | 15 -> 11-> 2                        |
| 5      | GEFC               | 17 -> 16-> 12-> 2                   |
| 6      | HC                 | 13 -> 2                             |
| 7      | HDEB               | 22 -> 20 -> 18 -> 3                 |
| 8      | GEB                | 21 -> 18 -> 3                       |
| 9      | GDFB               | 25 -> 23 -> 19 -> 3                 |
| 10     | HFB                | 24 -> 19 -> 3                       |

in MIN) of the network. Table 6 shows  $24 \times 24$  connection matrix for the network shown in Fig. 4 as there are 24 nodes/switches available in 4DGIN-1 design.

Any row having '1' in any column of the Connection matrix indicates that two nodes are interconnected by a link. For example 5th row of Table has value '1' in the columns

named 13, 15 and 19 which interprets that the node/switch 5 is connected with switch 13, switch 15 and switch 19.

In this paper reliability of 4DGIN-1 and other MINs are evaluated using efficient path based methods. Reliability is calculated using this efficient methodology which first enumerate minimal path sets and then evaluate reliability expression in compact of sum of disjoint products (SDP) form using multi-variable inversion (MVI) algorithm (Chaturvedi and Misra 2002).

Reliability values are calculated by assuming different switch reliability values. The terminal reliability values, evaluated for various MINs, are shown in Table 7.

The algorithms are coded in MATLAB 9.0 and implemented on a computer with a 3.00 GHz Intel (R) Core (TM) 2 Duo processor and 2 GB of RAM with Windows 7 operating system. After applying algorithm for the 4DGIN-1 we get,

Now consider the  $8 \times 8$  4DGIN-1 for the flow of data from source node 0 (000) wants to send data to destination node 6 (110). Network (Nodes/Links) =  $24/72$  (Number of switches used is 24 and Links = 72 to connect inputs and outputs)

**Table 6** Connection matrix for the network in Fig. 2

| Nodes/switches | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|----------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1              | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 2              | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 3              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 4              | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 5              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  |
| 6              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |
| 7              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 8              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  |
| 9              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 0  |
| 10             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  |
| 11             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 0  |
| 12             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  |
| 13             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 0  |
| 14             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 0  |
| 15             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  |
| 16             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  |
| 17             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  |
| 18             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  |
| 19             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  |
| 20             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  |
| 21             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 22             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 23             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 24             | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

## 5.2 Terminal reliability

Number of Redundant Paths for source–destination pair (source **0** to destination **6**) = 5

Number of disjoint Paths for any source–destination pair = 4

Enumerated minimal path set:

1 5 19 24

1 6 20 24

1 7 17 24

1 12 18 24

1 12 20 24

The terminal reliability equation is calculated by MVI algorithm =  $5r^4 - 2r^5 - 7r^6 + 4r^7 + 3r^8 - 2r^9$

System reliability = 0.8091813 when switch reliability ( $r$ ) = 0.90.

The terminal reliability for different tag values is computed by generating all possible paths for each source and destination pair of the network and then evaluating terminal reliability from the minimal paths assuming different switch reliability values. The terminal reliability values, evaluated for various MINs are shown in Table 7.

## 5.3 Broadcast reliability

Number of redundant paths for source–destination pair (source **0** to all the destinations) = 14

Enumerated minimal path set:

Number of redundant paths (14)

1-5-12-14-19-21-22-23-24

1-5-12-13-20-21-22-23-24

1-5-6-14-19-21-22-23-24

1-5-6-13-20-21-22-23-24

1-5-12-15-18-21-22-23-24

1-5-13-19-21-22-23-24

1-6-7-16-17-21-22-23-24

1-6-7-13-20-21-22-23-24

1-6-12-16-18-21-22-23-24

1-6-14-20-21-22-23-24

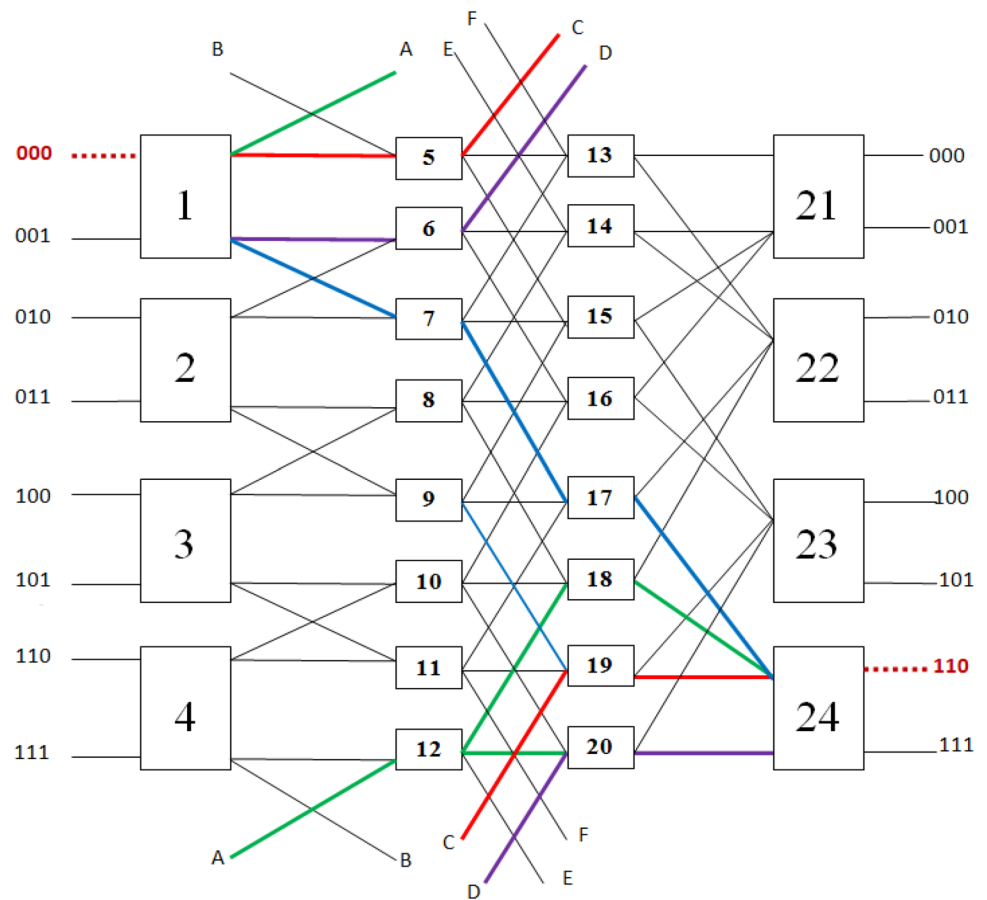
1- 7-12-13-20-21-22-23-24

1-7-12-15-18-21-22-23-24

1-7-15-17 -21-22-23-24

1-12-14-20-21-22-23-24

**Fig. 4** 4DGIN-1 showing four disjoint paths and five redundant paths for S-D node pair (0, 6;  $\delta = 6$ )



System reliability = 0.5889279 when switch reliability ( $r$ ) = 0.90

#### 5.4 Global or all terminal reliability (ATR)

Number of redundant paths for source–destination pair (all sources to all destinations) = 115

Number of redundant paths (115)

|                                        |                                           |                                                 |
|----------------------------------------|-------------------------------------------|-------------------------------------------------|
| 1 2 3 4 9 12 14 15 17 20 21 22 23 24   | 1 2 3 4 6 8 11 12 13 16 18 20 21 22 23 24 | 1 2 3 4 5 6 7 9 10 13 15 18 20 21 22 23 24      |
| 1 2 3 4 8 9 12 14 19 20 21 22 23 24    | 1 2 3 4 5 6 8 10 13 16 18 20 21 22 23 24  | 1 2 3 4 5 7 8 10 13 14 15 18 19 20 21 22 23 24  |
| 1 2 3 4 8 9 12 14 15 18 20 21 22 23 24 | 1 2 3 4 5 9 12 13 15 17 20 21 22 23 24    | 1 2 3 4 7 9 10 12 13 14 15 18 19 20 21 22 23 24 |
| 1 2 3 4 8 9 12 14 16 17 20 21 22 23 24 | 1 2 3 4 6 7 11 13 16 17 19 21 22 23 24    | 1 2 3 4 6 9 11 12 13 15 16 17 18 20 21 22 23 24 |
| 1 2 3 4 8 12 14 16 18 20 21 22 23 24   | 1 2 3 4 6 9 10 12 14 16 18 19 21 22 23 24 | 1 2 3 4 5 7 8 10 13 14 16 17 19 20 21 22 23 24  |
| 1 2 3 4 6 10 14 16 18 20 21 22 23 24   | 1 2 3 4 7 8 10 12 14 15 17 20 21 22 23 24 | 1 2 3 4 5 6 8 11 14 15 16 17 20 21 22 23 24     |
| 1 2 3 4 6 10 11 13 14 20 21 22 23 24   | 1 2 3 4 5 7 9 10 13 16 17 19 21 22 23 24  | 1 2 3 4 6 9 11 12 13 16 17 18 19 21 22 23 24    |
| 1 2 3 4 5 9 12 14 15 17 19 21 22 23 24 | 1 2 3 4 5 6 9 10 13 16 17 20 21 22 23 24  | 1 2 3 4 7 9 10 12 13 16 18 19 20 21 22 23 24    |
| 1 2 3 4 5 8 9 12 14 19 21 22 23 24     | 1 2 3 4 6 9 10 12 16 17 18 21 22 23 24    | 1 2 3 4 5 6 8 11 13 15 16 17 18 20 21 22 23 24  |
| 1 2 3 4 5 8 12 14 16 18 19 21 22 23 24 | 1 2 3 4 5 7 9 10 13 15 18 19 21 22 23 24  | 1 2 3 4 5 7 8 11 12 14 16 17 19 21 22 23 24     |
| 1 2 3 4 5 8 9 13 15 18 19 21 22 23 24  | 1 2 3 4 6 8 11 12 13 16 18 19 21 22 23 24 | 1 2 3 4 6 9 11 12 13 14 16 18 19 21 22 23 24    |
| 1 2 3 4 7 10 11 15 16 17 21 22 23 24   | 1 2 3 4 7 9 11 12 13 15 17 20 21 22 23 24 | 1 2 3 4 7 8 9 11 12 13 15 18 20 21 22 23 24     |
| 1 2 3 4 6 8 11 12 14 19 20 21 22 23 24 | 1 2 3 4 5 6 9 10 14 15 18 20 21 22 23 24  | 1 2 3 4 5 7 8 10 11 14 15 17 20 21 22 23 24     |
| 1 2 3 4 6 7 10 11 13 20 21 22 23 24    | 1 2 3 4 7 9 11 12 13 19 20 21 22 23 24    | 1 2 3 4 7 8 10 12 14 16 17 20 21 22 23 24       |
| 1 2 3 4 5 7 9 11 13 19 21 22 23 24     | 1 2 3 4 5 6 9 10 13 15 17 20 21 22 23 24  | 1 2 3 4 5 6 8 11 14 15 16 17 18 19 21 22 23 24  |
| 1 2 3 4 5 8 9 13 16 17 19 21 22 23 24  | 1 2 3 4 5 6 9 10 13 16 17 19 21 22 23 24  | 1 2 3 4 5 6 8 9 10 13 15 18 20 21 22 23 24      |
| 1 2 3 4 5 8 9 12 15 18 21 22 23 24     | 1 2 3 4 5 6 9 11 14 15 17 19 21 22 23 24  | 1 2 3 4 5 6 9 10 12 14 16 17 19 21 22 23 24     |
| 1 2 3 4 6 7 10 13 16 18 20 21 22 23 24 | 1 2 3 4 7 10 11 13 15 17 20 21 22 23 24   | 1 2 3 4 6 7 9 10 12 13 16 18 19 21 22 23 24     |

**Table 7** Multistage interconnection networks terminal reliability comparison

| S. no | Network types                                                                                                                                                                                                                                                                  | Network nodes/<br>links | Number of<br>paths | Disjoint paths | Node reliability ( $r = 0.99$ ) |           |              | Node reliability ( $r = 0.90$ ) |           |              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------|----------------|---------------------------------|-----------|--------------|---------------------------------|-----------|--------------|
|       |                                                                                                                                                                                                                                                                                |                         |                    |                | Two-terminal                    | Broadcast | All-terminal | Two-terminal                    | Broadcast | All-terminal |
| 1.    | Banyan (Jiang et al. 2008), Reverse Baseline (Yang and Wang 2005), Omega (Shen et al. 2001), Reverse Omega (Yang and Wang 2005), Delta (He and Delgado-Frias 2007), Shuffle exchange (SEN) (Kumar and Reddy 1987), Cube (Fan and Bruck 2000) and baseline (Yang and Wang 2005) | 12/32                   | 1                  | 1              | 0.9703                          | 0.9320    | 0.8864       | 0.7290                          | 0.4783    | 0.2824       |
| 2.    | Extra group network (Wei 1988)                                                                                                                                                                                                                                                 | 12/84                   | 3                  | 3              | 0.9797                          | 0.9133    | 0.8510       | 0.7770                          | 0.3468    | 0.1411       |
| 3.    | Augmented baseline network (ABN) (Aggarwal and Kaur 2009)                                                                                                                                                                                                                      | 8/76                    | 8                  | 2              | 0.9798                          | 0.9122    | 0.8502       | 0.7766                          | 0.3382    | 0.1613       |
| 4.    | Modified Baseline (Danilewicz and Rajewski 2014)                                                                                                                                                                                                                               | 48/192                  | 1                  | 1              | 0.9605                          | 0.7936    | 0.6173       | 0.6561                          | 0.0886    | 0.00636      |
| 5.    | Butterfly (Balkan et al. 2009) (Nieminen 2014)                                                                                                                                                                                                                                 | 32/48                   | 1                  | 1              | 0.9605                          | 0.8600    | 0.7249       | 0.6561                          | 0.2059    | 0.0344       |
| 6.    | Benes (Raponi et al. 2013)                                                                                                                                                                                                                                                     | 20/32                   | 4                  | 2              | 0.9797                          | 0.9503    | 0.3866       | 0.7782                          | 0.5523    | 0.9217       |
| 7.    | Clos network (Pham et al. 2013)                                                                                                                                                                                                                                                | 12/32                   | 4                  | 4              | 0.9800                          | 0.9509    | 0.9228       | 0.8099                          | 0.5904    | 0.4304       |
| 8.    | PHI (Sengupta and Bansal 2001)                                                                                                                                                                                                                                                 | 10/30                   | 5                  | 2              | 0.9787                          | 0.9105    | 0.8464       | 0.6665                          | 0.2984    | 0.1219       |
| 9.    | Extra stage (SEN+) (Kumar and Reddy 1987), extra stage cube (Yang and Wang 2004)                                                                                                                                                                                               | 16/40                   | 2                  | 2              | 0.9797                          | 0.9503    | 0.9216       | 0.7807                          | 0.5548    | 0.3887       |

Table 7 continued

| S. no | Network types                                                                                       | Network nodes/<br>links | Number of<br>paths | Disjoint paths | Node reliability (r) = 0.99 |           |              | Node reliability (r) = 0.90 |           |              |
|-------|-----------------------------------------------------------------------------------------------------|-------------------------|--------------------|----------------|-----------------------------|-----------|--------------|-----------------------------|-----------|--------------|
|       |                                                                                                     |                         |                    |                | Two-terminal                | Broadcast | All-terminal | Two-terminal                | Broadcast | All-terminal |
| 10.   | Augmented SEN (Kumar and Reddy 1987)                                                                | 24/144                  | 4                  | 2              | 0.9794                      | 0.9118    | 0.8490       | 0.7480                      | 0.3246    | 0.1406       |
| 11.   | Generalized SEN (Chiuyuan and Jing-Kai 2006)                                                        | 20/50                   | 2                  | 2              | 0.9797                      | 0.9042    | 0.8179       | 0.7807                      | 0.3452    | 0.1216       |
| 12.   | Dot network (Bansal et al. 1992)                                                                    | 13/34                   | 1                  | 1              | 0.9500                      | 0.9135    | 0.8775       | 0.5904                      | 0.3874    | 0.2542       |
| 13.   | Theta (Sengupta and Bansal 2004)                                                                    | 10/28                   | 3                  | 1              | 0.9600                      | 0.8594    | 0.7528       | 0.6223                      | 0.1921    | 0.0422       |
| 14.   | Enhanced IADM (Chen and Chung 2001)                                                                 | 32/112                  | 7                  | 3              | 0.9800                      | 0.9135    | 0.8514       | 0.8082                      | 0.3864    | 0.1823       |
| 15.   | Gamma (GIN) (Parker and Raghavendra 1984)                                                           | 32/88                   | 1                  | 1              | 0.9605                      | 0.8948    | 0.7249       | 0.6561                      | 0.2965    | 0.0344       |
| 16.   | Extra stage GIN (Lee and Hegazy 1998)                                                               | 40/112                  | 8                  | 2              | 0.9799                      | 0.9131    | 0.8487       | 0.7979                      | 0.3662    | 0.1364       |
| 17.   | Balanced gamma (Cheng 2006)                                                                         | 32/96                   | 4                  | 2              | 0.9799                      | 0.8950    | 0.8504       | 0.7938                      | 0.3014    | 0.1654       |
| 18.   | Reliable Interconnection Network-2 (RIN-2) (Rajkumar and Goyal 2015d)                               | 28/80                   | 6                  | 4              | 0.9899                      | 0.9896    | 0.9602       | 0.8859                      | 0.8476    | 0.5980       |
| 19.   | Monogamma GIN (Chuang 1996)                                                                         | 32/88                   | 2                  | 1              | 0.9798                      | 0.8944    | 0.7845       | 0.7217                      | 0.2882    | 0.0699       |
| 20.   | Partial chained GIN (Chen et al. 2000)                                                              | 32/88                   | 18                 | 2              | 0.9798                      | 0.9130    | 0.8511       | 0.7844                      | 0.3615    | 0.1778       |
| 21.   | Fully chained GIN (Chen et al. 2000)                                                                | 32/112                  | 64                 | 2              | 0.9788                      | 0.9128    | 0.8511       | 0.7791                      | 0.3525    | 0.1766       |
| 22.   | No-Backtracking (NBGIN) (Chen and Chung 2001), Combining switches MIN (CSMIN) (Chen and Chung 2005) | 28/80                   | 4                  | 2              | 0.9799                      | 0.9126    | 0.8839       | 0.7873                      | 0.3527    | 0.2236       |



**Table 7** continued

| S. no | Network types                                                                       | Network nodes/<br>links | Number of<br>paths | Disjoint paths | Node reliability (r) = 0.99 |           |              | Node reliability (r) = 0.90 |           |              |
|-------|-------------------------------------------------------------------------------------|-------------------------|--------------------|----------------|-----------------------------|-----------|--------------|-----------------------------|-----------|--------------|
|       |                                                                                     |                         |                    |                | Two-terminal                | Broadcast | All-terminal | Two-terminal                | Broadcast | All-terminal |
| 23.   | 3-disjoint paths GIN (3DGIN) (Chen et al. 2003)                                     | 28/80                   | 4                  | 3              | 0.9800                      | 0.9134    | 0.8862       | 0.8057                      | 0.3765    | 0.2623       |
| 24.   | Minimal links traversed 32/96 dynamic rerouting network (MinGIN) (Chen and Fu 2004) | 32/96                   | 4                  | 2              | 0.9797                      | 0.9126    | 0.8491       | 0.7873                      | 0.3527    | 0.1467       |
| 25.   | Incomplete GIN (Chen 2013)                                                          | 50/140                  | 6                  | 2              | 0.9799                      | 0.8948    | 0.8139       | 0.7979                      | 0.2946    | 0.0796       |
| 26.   | 4-disjoint paths GIN (4DGIN) (Rajkumar and Goyal 2014)                              | 24/72                   | 7                  | 4              | 0.9801                      | 0.9509    | 0.9225       | 0.8093                      | 0.5889    | 0.4269       |
| 27.   | Reliable Interconnection Network-1 (RIN-1) (Rajkumar and Goyal 2015d)               | 32/104                  | 12                 | 4              | 0.9998                      | 0.9995    | 0.9991       | 0.9776                      | 0.9419    | 0.9082       |
| 28.   | Fault Tolerant Interconnection Network (FTIN) (Rajkumar and Goyal 2015b)            | 24/72                   | 6                  | 4              | 0.9801                      | 0.9510    | 0.9227       | 0.8095                      | 0.5896    | 0.4283       |

|                                        |                                           |                                                 |
|----------------------------------------|-------------------------------------------|-------------------------------------------------|
| 1 2 3 4 6 7 11 13 19 20 21 22 23 24    | 1 2 3 4 6 8 11 12 14 16 18 19 21 22 23 24 | 1 2 3 4 5 6 9 11 13 15 17 20 21 22 23 24        |
| 1 2 3 4 7 11 13 15 17 19 21 22 23 24   | 1 2 3 4 5 8 12 13 16 18 20 21 22 23 24    | 1 2 3 4 6 7 8 11 12 14 16 17 19 21 22 23 24     |
| 1 2 3 4 7 10 15 16 17 18 21 22 23 24   | 1 2 3 4 5 7 8 11 13 15 18 19 21 22 23 24  | 1 2 3 4 6 9 10 12 14 15 18 20 21 22 23 24       |
| 1 2 3 4 6 11 13 14 19 20 21 22 23 24   | 1 2 3 4 7 8 11 12 13 16 18 20 21 22 23 24 | 1 2 3 4 6 7 9 11 12 13 16 18 19 21 22 23 24     |
| 1 2 3 4 5 7 9 10 15 17 18 21 22 23 24  | 1 2 3 4 7 8 11 12 13 15 18 19 21 22 23 24 | 1 2 3 4 6 9 10 12 14 16 17 20 21 22 23 24       |
| 1 2 3 4 5 8 13 16 18 19 21 22 23 24    | 1 2 3 4 7 8 11 12 13 16 17 20 21 22 23 24 | 1 2 3 4 7 8 11 12 14 16 17 20 21 22 23 24       |
| 1 2 3 4 5 6 8 10 13 14 20 21 22 23 24  | 1 2 3 4 5 6 9 10 14 15 18 19 21 22 23 24  | 1 2 3 4 5 7 8 10 13 14 15 17 20 21 22 23 24     |
| 1 2 3 4 6 8 10 12 16 18 21 22 23 24    | 1 2 3 4 5 6 9 11 13 16 17 19 21 22 23 24  | 1 2 3 4 5 6 8 10 11 14 15 17 20 21 22 23 24     |
| 1 2 3 4 7 8 10 12 15 16 18 21 22 23 24 | 1 2 3 4 7 9 11 12 14 15 17 19 21 22 23 24 | 1 2 3 4 7 9 10 12 13 15 16 18 19 21 22 23 24    |
| 1 2 3 4 5 8 12 15 16 18 21 22 23 24    | 1 2 3 4 7 9 11 12 13 15 18 19 21 22 23 24 | 1 2 3 4 7 9 10 12 13 14 16 17 19 20 21 22 23 24 |
| 1 2 3 4 5 9 13 15 17 19 21 22 23 24    | 1 2 3 4 5 7 9 10 13 15 17 20 21 22 23 24  | 1 2 3 4 7 8 10 12 13 16 18 20 21 22 23 24       |
| 1 2 3 4 5 9 12 15 17 18 21 22 23 24    | 1 2 3 4 6 10 11 14 16 17 20 21 22 23 24   | 1 2 3 4 7 8 11 12 14 15 17 19 21 22 23 24       |
| 1 2 3 4 5 7 8 11 15 16 17 21 22 23 24  | 1 2 3 4 5 7 8 11 14 15 17 19 21 22 23 24  | 1 2 3 4 6 8 11 12 14 16 17 20 21 22 23 24       |
| 1 2 3 4 5 7 9 11 15 17 21 22 23 24     | 1 2 3 4 5 6 9 11 14 15 17 20 21 22 23 24  | 1 2 3 4 5 6 8 11 14 15 17 19 20 21 22 23 24     |
| 1 2 3 4 6 8 10 12 14 20 21 22 23 24    | 1 2 3 4 5 6 9 10 14 16 18 19 21 22 23 24  | 1 2 3 4 6 9 11 12 14 15 16 17 18 19 21 22 23 24 |
| 1 2 3 4 6 7 10 16 17 18 21 22 23 24    | 1 2 3 4 5 8 9 12 13 16 17 20 21 22 23 24  | 1 2 3 4 5 6 7 8 10 11 14 15 18 19 21 22 23 24   |
| 1 2 3 4 5 6 9 11 13 14 19 21 22 23 24  | 1 2 3 4 5 6 8 10 14 15 18 20 21 22 23 24  | 1 2 3 4 5 6 9 10 11 14 16 17 19 21 22 23 24     |
| 1 2 3 4 6 7 10 11 16 17 21 22 23 24    | 1 2 3 4 7 8 11 12 14 15 18 19 21 22 23 24 | 1 2 3 4 5 7 8 10 14 15 17 18 20 21 22 23 24     |
| 1 2 3 4 5 8 9 13 14 19 21 22 23 24     | 1 2 3 4 5 6 8 10 14 16 18 19 21 22 23 24  | 1 2 3 4 7 8 10 11 12 13 15 18 20 21 22 23 24    |
|                                        | 1 2 3 4 5 7 9 10 13 16 18 19 21 22 23 24  | 1 2 3 4 5 7 8 11 13 16 17 19 21 22 23 24        |
|                                        | 1 2 3 4 7 8 10 12 14 15 18 20 21 22 23 24 | 1 2 3 4 6 7 9 10 11 12 14 15 18 19 21 22 23 24  |

System reliability = 0.4267499 when switch (node) reliability ( $r$ ) = 0.90

### 5.5 Multi-source-multi terminal reliability

The probability of successful communication between given set of multiple source nodes (S) and multiple terminal nodes (T) in the network.

**Benes Network** (Raponi et al. 2013) is a rearrangeable non-blocking MIN due to the combination of baseline and reverse baseline network where the intermediate stages are overlapped. Network designed by using  $(2(\log_2 N) - 1)$  stages numbered from stage 0 to stage  $n$  where  $n = \log_2 N$ . Each stage is composed of  $(N/2)$  switching elements. The switching elements used  $2 \times 2$  size crossbars for all the stages.

Now consider the  $8 \times 8$  Benes Network as shown in Fig. 5 for the flow of data from multiple sources to multiple destinations.

#### Case 1: 3S/4T

Source nodes = 0, 1, and 2 (any 3 sources)

Destination nodes = 4, 5, 6, and 7 (any 4 destinations)

Number of redundant paths for source–destination pair (3S/4T) = 4

Enumerated minimal path set:

1 2 7 12 16 19 20  
1 2 7 11 16 19 20  
1 2 5 10 14 19 20  
1 2 5 9 14 19 20

System reliability = 0.8064008 when switch reliability ( $r$ ) = 0.95

#### Case 2: 1S/6T

Source nodes = 0 (any 1 source)

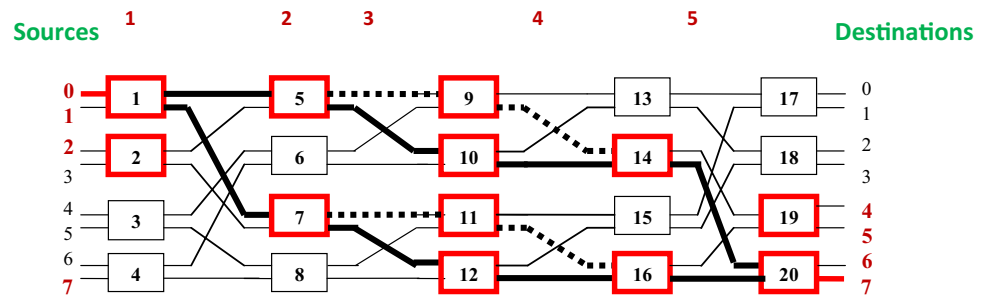
Destination nodes = 2, 3, 4, 5, 6, and 7 (any 6 destinations)

Number of redundant paths for source–destination pair (1S/6T) = 12

Number of redundant paths (12)

|                            |                            |                            |
|----------------------------|----------------------------|----------------------------|
| 1-7-12-15-16-18-19-20      | 1-5-7-9-11-13-16-18-19-20  | 1-5-9-13-14-18-19-20       |
| 1-5-7-10-12-14-15-18-19-20 | 1-5-7-9-12-13-16-18-19-20  | 1-5-10-13-14-18-19-20      |
| 1-5-7-9-12-14-15-18-19-20  | 1-5-7-10-11-13-16-18-19-20 | 1-5-7-9-11-14-15-18-19-20  |
| 1-7-11-15-16-18-19-20      | 1-5-7-10-12-13-16-18-19-20 | 1-5-7-10-11-14-15-18-19-20 |

**Fig. 5**  $8 \times 8$  Benes network showing two disjoint paths and four redundant paths for case 1



**Table 8** Multi source (s) multi terminal (t) reliability comparison (node reliability = 0.95)

| No. | Network types                                                                                                                                                                                                                                                                  | 1S/2D  | 2S/3D  | 3S/4D  | 4S/5D  | 5S/6D  | 6S/7D  | 7S/8D  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
| 1.  | Banyan (Jiang et al. 2008), Reverse Baseline (Yang and Wang 2005), Omega (Shen et al. 2001), Reverse Omega (Yang and Wang 2005), Delta (He and Delgado-Frias 2007), Shuffle exchange (SEN) (Kumar and Reddy 1987), Cube (Fan and Bruck 2000) and Baseline (Yang and Wang 2005) | 0.8574 | 0.8145 | 0.7351 | 0.6302 | 0.5987 | 0.5688 | 0.5403 |
| 2.  | Extra Group Network (Wei 1988)                                                                                                                                                                                                                                                 | 0.8490 | 0.7535 | 0.6709 | 0.6007 | 0.5421 | 0.4893 | 0.4415 |
| 3.  | ABN (Aggarwal and Kaur 2009)                                                                                                                                                                                                                                                   | 0.8403 | 0.7473 | 0.6745 | 0.6022 | 0.5379 | 0.4782 | 0.4316 |
| 4.  | Modified baseline (Danilewicz and Rajewski 2014)                                                                                                                                                                                                                               | 0.8145 | 0.4632 | 0.3774 | 0.3074 | 0.2774 | 0.2259 | 0.2039 |
| 5.  | Butterfly (Balkan et al. 2009; Nieminen 2014)                                                                                                                                                                                                                                  | 0.7350 | 0.4632 | 0.3405 | 0.3073 | 0.2503 | 0.2259 | 0.2039 |
| 6.  | Benes (Raponi et al. 2013)                                                                                                                                                                                                                                                     | 0.8935 | 0.8488 | 0.8064 | 0.7606 | 0.7146 | 0.6789 | 0.6449 |
| 7.  | Clos network (Pham et al. 2013)                                                                                                                                                                                                                                                | 0.9024 | 0.8573 | 0.8145 | 0.7738 | 0.7351 | 0.6983 | 0.6634 |
| 8.  | SEN+ (Kumar and Reddy 1987) Extra stage cube (Yang and Wang 2004)                                                                                                                                                                                                              | 0.8939 | 0.8492 | 0.8067 | 0.7611 | 0.7152 | 0.6794 | 0.6454 |
| 9.  | PHI (Sengupta and Bansal 2001)                                                                                                                                                                                                                                                 | 0.8209 | 0.6939 | 0.6160 | 0.5559 | 0.5017 | 0.4338 | 0.3915 |
| 10. | ASEN (Kumar and Reddy 1987)                                                                                                                                                                                                                                                    | 0.8316 | 0.7452 | 0.6677 | 0.5916 | 0.5339 | 0.4783 | 0.4315 |
| 11. | Generalized SEN (Chiuyuan and Jing-Kai 2006)                                                                                                                                                                                                                                   | 0.8145 | 0.7737 | 0.6983 | 0.5987 | 0.4876 | 0.4632 | 0.4401 |
| 12. | Dot network (Bansal et al. 1992)                                                                                                                                                                                                                                               | 0.7737 | 0.7350 | 0.6302 | 0.5987 | 0.5688 | 0.5403 | 0.5133 |
| 13. | Theta (Sengupta and Bansal 2004)                                                                                                                                                                                                                                               | 0.6571 | 0.4702 | 0.4244 | 0.3804 | 0.3433 | 0.2533 | 0.2286 |
| 14. | Enhanced IADM (Chen and Chung 2001)                                                                                                                                                                                                                                            | 0.8572 | 0.7733 | 0.6978 | 0.6295 | 0.5681 | 0.5125 | 0.4625 |
| 15. | Gamma (Parker and Raghavendra 1984)                                                                                                                                                                                                                                            | 0.7682 | 0.6287 | 0.5121 | 0.4181 | 0.3405 | 0.2774 | 0.2259 |
| 16. | Extra Stage GIN (Lee and Hegazy 1998)                                                                                                                                                                                                                                          | 0.8543 | 0.7665 | 0.6860 | 0.6128 | 0.5463 | 0.4861 | 0.4319 |
| 17. | Balanced Gamma Network (Cheng et al. 2006)                                                                                                                                                                                                                                     | 0.7699 | 0.6271 | 0.5095 | 0.4587 | 0.4139 | 0.3736 | 0.3043 |
| 18. | RIN-2 (Rajkumar and Goyal 2015d)                                                                                                                                                                                                                                               | 0.9470 | 0.9438 | 0.8959 | 0.8927 | 0.8469 | 0.8437 | 0.8002 |
| 19. | Monogamma GIN (Chuang 1996)                                                                                                                                                                                                                                                    | 0.8049 | 0.6527 | 0.5294 | 0.4538 | 0.4095 | 0.3770 | 0.2971 |
| 20. | Partial chained GIN (Chen et al. 2000)                                                                                                                                                                                                                                         | 0.8501 | 0.7672 | 0.6871 | 0.5891 | 0.5570 | 0.5052 | 0.4563 |
| 21. | Fully chained GIN (Chen et al. 2000)                                                                                                                                                                                                                                           | 0.8471 | 0.7539 | 0.6824 | 0.6114 | 0.5548 | 0.5022 | 0.4557 |
| 22. | NBGIN (Chen and Chung 2001), (CSMIN) (Chen and Chung 2005)                                                                                                                                                                                                                     | 0.8960 | 0.7989 | 0.7505 | 0.6727 | 0.6306 | 0.5691 | 0.5339 |
| 23. | 3DGIN (Chen et al. 2003)                                                                                                                                                                                                                                                       | 0.9018 | 0.8138 | 0.7719 | 0.6600 | 0.6254 | 0.5934 | 0.5629 |
| 24. | MinGIN (Chen and Fu 2004)                                                                                                                                                                                                                                                      | 0.8471 | 0.7539 | 0.6774 | 0.6031 | 0.5372 | 0.4849 | 0.4348 |
| 25. | Incomplete GIN (Chen 2013)                                                                                                                                                                                                                                                     | 0.8545 | 0.7673 | 0.6873 | 0.6145 | 0.5479 | 0.4875 | 0.4329 |
| 26. | 4DGIN (Rajkumar and Goyal 2014)                                                                                                                                                                                                                                                | 0.9025 | 0.8572 | 0.8143 | 0.7735 | 0.7348 | 0.6979 | 0.6630 |
| 27. | FTIN (Rajkumar and Goyal 2015b)                                                                                                                                                                                                                                                | 0.9025 | 0.8573 | 0.8144 | 0.7736 | 0.7349 | 0.6982 | 0.6632 |
| 28. | RIN-1 (Rajkumar and Goyal 2015d)                                                                                                                                                                                                                                               | 0.9948 | 0.9893 | 0.9868 | 0.9814 | 0.9800 | 0.9792 | 0.9786 |

System reliability = 0.8007364 when switch reliability (r) = 0.95

The above mentioned approach enumerated minimal paths for 2-TR, broadcast and ATR networks and

successfully evaluates reliability of MINs with multiple-sources and multiple-destinations nodes.

Thus a variety of MINs and their reliability parameters are evaluated and shown in Tables 7, 8, 9.

**Table 9** Multi source (s) multi terminal (t) reliability comparison (switch reliability  $r = 0.5$ )

| No. | Network types                                                                                                                                                                                                                                                                  | 1S/2D  | 2S/3D                | 3S/4D                 | 4S/5D                 | 5S/6D                 | 6S/7D                 | 7S/8D                 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1.  | Banyan (Jiang et al. 2008), Reverse Baseline (Yang and Wang 2005), Omega (Shen et al. 2001), Reverse Omega (Yang and Wang 2005), Delta (He and Delgado-Frias 2007), Shuffle Exchange (SEN) (Kumar and Reddy 1987), Cube (Fan and Bruck 2000) and Baseline (Yang and Wang 2005) | 0.125  | 0.0625               | 0.0156                | 0.0020                | $9.7 \times 10^{-4}$  | $4.8 \times 10^{-4}$  | $2.44 \times 10^{-4}$ |
| 2.  | Extra Group Network (Wei 1988)                                                                                                                                                                                                                                                 | 0.0120 | $2.1 \times 10^{-4}$ | $1.3 \times 10^{-5}$  | $1.6 \times 10^{-6}$  | $3.9 \times 10^{-7}$  | $9.9 \times 10^{-8}$  | $2.48 \times 10^{-8}$ |
| 3.  | ABN (Aggarwal and Kaur 2009)                                                                                                                                                                                                                                                   | 0.0300 | 0.0032               | $7.95 \times 10^{-4}$ | $7.62 \times 10^{-5}$ | $1.9 \times 10^{-5}$  | $2.8 \times 10^{-6}$  | $7 \times 10^{-7}$    |
| 4.  | Modified baseline (Danilewicz and Rajewski 2014)                                                                                                                                                                                                                               | 0.0625 | $3 \times 10^{-5}$   | $1.9 \times 10^{-6}$  | $1.1 \times 10^{-7}$  | $2.9 \times 10^{-8}$  | $1.8 \times 10^{-9}$  | $4.6 \times 10^{-10}$ |
| 5.  | Butterfly (Balkan et al. 2009; Nieminen 2014)                                                                                                                                                                                                                                  | 0.016  | $3 \times 10^{-5}$   | $4.7 \times 10^{-7}$  | $1.1 \times 10^{-7}$  | $7.4 \times 10^{-9}$  | $1.8 \times 10^{-9}$  | $4.6 \times 10^{-10}$ |
| 6.  | Benes (Raponi et al. 2013)                                                                                                                                                                                                                                                     | 0.085  | 0.042                | 0.021                 | 0.006                 | 0.0016                | $7.8 \times 10^{-4}$  | $3.9 \times 10^{-4}$  |
| 7.  | Clos Network (Pham et al. 2013)                                                                                                                                                                                                                                                | 0.234  | 0.1171               | 0.0586                | 0.0293                | 0.015                 | 0.007                 | 0.003                 |
| 8.  | SEN+ (Kumar and Reddy 1987)                                                                                                                                                                                                                                                    | 0.109  | 0.054                | 0.027                 | 0.008                 | 0.002                 | 0.001                 | $5.3 \times 10^{-4}$  |
| 9.  | Extra Stage Cube (Yang and Wang 2004)                                                                                                                                                                                                                                          |        |                      |                       |                       |                       |                       |                       |
| 10. | PHI (Sengupta and Bansal 2001)                                                                                                                                                                                                                                                 | 0.005  | $2.9 \times 10^{-4}$ | $2.6 \times 10^{-5}$  | $6.6 \times 10^{-6}$  | $1.6 \times 10^{-6}$  | $1.5 \times 10^{-7}$  | $3.8 \times 10^{-8}$  |
| 11. | ASEN (Kumar and Reddy 1987)                                                                                                                                                                                                                                                    | 0.0192 | 0.0016               | $4 \times 10^{-4}$    | $3.7 \times 10^{-5}$  | $4.6 \times 10^{-6}$  | $4 \times 10^{-7}$    | $1 \times 10^{-7}$    |
| 12. | Generalized SEN (Chiu-Yuan and Jing-Kai 2006)                                                                                                                                                                                                                                  | 0.0625 | 0.0313               | 0.0078                | $9.7 \times 10^{-4}$  | $6.1 \times 10^{-5}$  | $3.05 \times 10^{-5}$ | $1.5 \times 10^{-5}$  |
| 13. | Dot network (Bansal et al. 1992)                                                                                                                                                                                                                                               | 0.031  | 0.015                | 0.002                 | $9.7 \times 10^{-4}$  | $4.8 \times 10^{-4}$  | $2.4 \times 10^{-4}$  | $1 \times 10^{-4}$    |
| 14. | Theta (Sengupta and Bansal 2004)                                                                                                                                                                                                                                               | 0.002  | $4.8 \times 10^{-6}$ | $1.2 \times 10^{-6}$  | $1.82 \times 10^{-7}$ | $4.5 \times 10^{-8}$  | $7.7 \times 10^{-10}$ | $1.9 \times 10^{-10}$ |
| 15. | Enhanced IADM (Chen and Chung 2001)                                                                                                                                                                                                                                            | 0.096  | 0.017                | 0.004                 | $7.5 \times 10^{-4}$  | $1.57 \times 10^{-4}$ | $3.24 \times 10^{-5}$ | $6.83 \times 10^{-6}$ |
| 16. | Gamma (Parker and Raghavendra 1984)                                                                                                                                                                                                                                            | 0.0156 | 0.0015               | $9.15 \times 10^{-5}$ | $7.63 \times 10^{-6}$ | $4.76 \times 10^{-7}$ | $2.98 \times 10^{-8}$ | $1.86 \times 10^{-9}$ |
| 17. | Extra stage GIN (Lee and Hegazy 1998)                                                                                                                                                                                                                                          | 0.0459 | 0.006                | $6.95 \times 10^{-4}$ | $7.5 \times 10^{-5}$  | $7.99 \times 10^{-6}$ | $7.45 \times 10^{-7}$ | $8.66 \times 10^{-8}$ |
| 18. | Balanced Gamma (Cheng et al. 2006)                                                                                                                                                                                                                                             | 0.018  | 0.0011               | $5.14 \times 10^{-5}$ | $9.6 \times 10^{-6}$  | $2.4 \times 10^{-6}$  | $6 \times 10^{-7}$    | $3.8 \times 10^{-8}$  |
| 19. | RIN-2 (Rajkumar and Goyal 2015d)                                                                                                                                                                                                                                               | 0.2188 | 0.1069               | 0.0385                | 0.0228                | 0.0070                | 0.0040                | 0.0014                |
| 20. | Monogamma GIN (Chuang 1996)                                                                                                                                                                                                                                                    | 0.023  | 0.001                | $6.3 \times 10^{-5}$  | $6.6 \times 10^{-6}$  | $1.6 \times 10^{-6}$  | $1.4 \times 10^{-7}$  | $1.1 \times 10^{-8}$  |
| 21. | Partial chained GIN (Chen et al. 2000)                                                                                                                                                                                                                                         | 0.031  | 0.007                | 0.0011                | $1.56 \times 10^{-4}$ | $5.1 \times 10^{-5}$  | $1.4 \times 10^{-5}$  | $4.5 \times 10^{-6}$  |
| 22. | Fully chained GIN (Chen et al. 2000)                                                                                                                                                                                                                                           | 0.026  | 0.001                | $3.9 \times 10^{-4}$  | $4.9 \times 10^{-5}$  | $2.01 \times 10^{-5}$ | $7.1 \times 10^{-6}$  | $1.14 \times 10^{-7}$ |
| 23. | NBGIN (Chen and Chung 2001), (CSMIN) (Chen and Chung 2005)                                                                                                                                                                                                                     | 0.125  | 0.01                 | 0.0027                | $4.3 \times 10^{-4}$  | $8.15 \times 10^{-5}$ | $2.04 \times 10^{-5}$ | $5.4 \times 10^{-6}$  |
| 24. | 3DGIN (Chen et al. 2003)                                                                                                                                                                                                                                                       | 0.1562 | 0.0344               | 0.0107                | $6.3 \times 10^{-4}$  | $1.7 \times 10^{-4}$  | $6.7 \times 10^{-5}$  | $2.7 \times 10^{-5}$  |
| 25. | MinGIN (Chen and Fu 2004)                                                                                                                                                                                                                                                      | 0.041  | 0.004                | $7.4 \times 10^{-4}$  | $6.6 \times 10^{-5}$  | $7.5 \times 10^{-6}$  | $1.8 \times 10^{-6}$  | $3.4 \times 10^{-7}$  |
| 26. | Incomplete GIN (Chen 2013)                                                                                                                                                                                                                                                     | 0.05   | 0.009                | 0.001                 | $1.2 \times 10^{-4}$  | $1.18 \times 10^{-5}$ | $1.13 \times 10^{-6}$ | $1.14 \times 10^{-7}$ |
| 27. | 4DGIN (Rajkumar and Goyal 2014)                                                                                                                                                                                                                                                | 0.1954 | 0.0815               | 0.0366                | 0.0137                | 0.0054                | 0.0022                | 0.001                 |
| 28. | FTIN (Rajkumar and Goyal 2015b)                                                                                                                                                                                                                                                | 0.1875 | 0.0760               | 0.0304                | 0.0152                | 0.0058                | 0.0029                | 0.0012                |
| 29. | RIN-1 (Rajkumar and Goyal 2015d)                                                                                                                                                                                                                                               | 0.3585 | 0.1134               | 0.0703                | 0.0266                | 0.0215                | 0.0188                | 0.0159                |

## 6 Conclusion

This paper has attempted to compare various MIN topologies reliability and concentrates more on reliability evaluation of MINs with multiple sources and multiple destinations nodes. The versatility of the approach is that it can be also used for evaluating all the network reliability parameters (2-TR, broadcast and ATR). Reliability of some recent and industrial used MIN topology is evaluated and all the network reliability parameters are compared here. Path Tracing Algorithm is applicable to trace minimal paths for any MINs. Thus complex MINs/clusters can also be investigated and compared detail in future.

## References

- Aggarwal R, Kaur L (2009) An efficient routing scheme to provide more fault-tolerance for an irregular multistage interconnection network'. *Advance Computing Conference, IEEE International*, pp 94–98
- Balkan AO, Qu G, Vishkin U (2009) 'Mesh-of-trees and alternative interconnection networks for single-chip parallelism', very large scale integration (VLSI) systems. *IEEE Trans* 17(10):1419–1432
- Bansal P, Singh K, Joshi RC (1992) 'Quad tree: a cost-effective fault-tolerant multistage interconnection network'. In: *Proceedings of Eleventh Annual Joint Conference of the IEEE Computer and Communications Societies, IEEE*, 2, pp 860–866
- Birolini A (2007) *Reliability engineering: theory and practice*. Springer, Berlin Heidelberg
- Blake JT, Trivedi KS (1989) Multistage interconnection network reliability. *IEEE Trans Comput* 38(11):1600–1604
- Bobbio A, Terruggia R, Ciancamerla E, Minichino M (2011) 'Reliability analysis of multi-source multi-sink critical interacting systems'. In: *Proceedings of 3rd International Workshop on Dependable Control of Discrete Systems*, pp 127–132
- Chandrashekar T, Goyal NK (2013) An approach to evaluate multiple node pair reliability for simultaneous capacity requirements. *Int J Perform Eng* 9(4):357–366
- Chaturvedi SK, Misra KB (2002) An efficient multi-variable inversion algorithm for reliability evaluation of complex systems using path sets. *Int J Reliab Qual Saf Eng* 9(3):237–259
- Chen Z (2013) 'A class of incomplete gamma interconnection network' [Report]. Tsinghua University, China, pp 1–12
- Chen C-W, Chung C-P (2001) Fault-tolerant gamma interconnection network without backtracking. *J Syst Softw* 58(1):23–31
- Chen C-W, Chung C-P (2005) Designing a disjoint paths interconnection network with fault tolerance and collision solving. *J Supercomput* 34(1):63–80
- Chen C-W, Fu S-C (2004) A minimal links traversed dynamic rerouting network. *Parallel Comput* 30(7):883–898
- Chen C-W, Lu N-P, Chen T-F, Chung C-P (2000) Fault-tolerant gamma interconnection networks by chaining. *IEEE Proc-Comput Digit Tech* 147(2):75–81
- Chen C-W, Lu N-P, Chung C-P (2003) 3-Disjoint gamma interconnection networks. *J Syst Softw* 66(2):129–134
- Cheng L, Venkatesan R, Heys HM (2006) 'Architecture and performance analysis of the multicast balanced gamma switch for broadband communications'. In: *Proceedings of IEEE International Conference on Computer Systems and Applications*, pp 381–388
- Chiuyuan C, Jing-Kai L (2006) An efficient tag-based routing algorithm for the backward network of a bidirectional general shuffle-exchange network. *IEEE Commun Lett* 10(4):296–298
- Chuang P-J (1996) CGIN: a fault tolerant modified gamma interconnection network. *IEEE Trans Parallel Distrib Syst* 7(12):1301–1306
- Danilewicz G, Rajewski R (2014) The architecture and strict-sense nonblocking conditions of a new baseline-based optical switching network composed of symmetrical and asymmetrical switching elements. *IEEE Trans Commun* 62(3):1058–1069
- Distefano S (2009) 'Reliability and dependability modeling and analysis of dynamic aspects in complex systems'. In: *Proceedings of IEEE International Conference on Dependable, autonomic and secure computing, Chengdu, China*, pp 43–48
- Fan CC, Bruck J (2000) Tolerating multiple faults in multistage interconnection networks with minimal extra stages. *IEEE Trans Comput* 49(9):998–1004
- Goyal NK (2006) On some aspects of reliability analysis and design of communication networks, Ph.D. dissertation, Indian Institute of Technology Kharagpur, India
- Goyal NK, Misra KB (2008) Optimum Link Capacity Allocation in a Communication Network. *IE (I) J-ET* 88:18–21
- Goyal NK, Misra RB, Chaturvedi SK (2005) SNEM: a new approach to evaluate terminal pair reliability of communication networks. *J Qual Maint Eng* 11(3):239–253
- Gunawan I (2008a) Redundant paths and reliability bounds in gamma networks. *Appl Math Model* 32(4):588–594
- Gunawan I (2008b) Reliability analysis of shuffle-exchange network systems. *Reliab Eng Syst Saf* 93(2):271–276
- Gunawan I (2013) Reliability prediction of distributed systems using Monte Carlo method. *Int J Reliab Saf* 7(3):235–248
- Gunawan I, Fard NS (2012) Terminal reliability assessment of gamma and extra-stage gamma networks. *Int J Qual Reliab Manag* 29(7):820–831
- He R, Delgado-Frias JG (2007) Fault tolerant interleaved switching fabrics for scalable high-performance routers. *IEEE Trans Parallel Distrib Syst* 18(12):1727–1739
- Jiang X, Pattavina A, Horiguchi S (2008) Strictly nonblocking f-cast photonic networks. *IEEE/ACM Trans Netw (TON)* 16(3):732–745
- Koren I, Mani Krishna C (2010) *Fault-tolerant systems*. Morgan Kaufmann Publishers, San Francisco
- Kumar V, Reddy S (1987) Augmented shuffle-exchange multistage interconnection networks. *IEEE Trans Comput* 20(6):30–40
- Lee KY, Hegazy W (1998) The extra stage gamma network. *IEEE Trans Comput* 37(11):1445–1450
- Li S-Y, Tan X (2009) On rearrangeability of tandem connection of banyan-type networks. *IEEE Trans Commun* 57(1):164–170
- Nieminen E (2014) A contention-free parallel access by butterfly networks for turbo interleavers. *IEEE Trans Inf Theory* 60(1):237–251
- Park Wei S, Gyungho L (1988) Extra group network: a cost-effective fault-tolerant multistage interconnection network. *ACM SIGARCH Comput Archit News* 16(2):108–115
- Parker D, Raghavendra C (1984) The gamma network. *IEEE Trans Comput* 100(4):367–373
- Pham P-H, Song J, Park J, Kim C (2013) 'Design and implementation of an on-chip permutation network for multiprocessor system-on-chip', very large scale integration (VLSI) systems. *IEEE Trans* 21(1):173–177
- Rajkumar S, Goyal NK (2014) Design of 4-disjoint gamma interconnection network layouts and reliability analysis of gamma interconnection networks. *J Supercomput* 69(1):468–491
- Rajkumar S, Goyal NK (2015a) Review of multistage interconnection networks reliability and fault-tolerance. *IETE Tech Review*. doi:10.1080/02564602.2015.1102098

- Rajkumar S, Goyal NK (2015b) Fault tolerant interconnection network design. *IETE Tech Rev*. doi:[10.1080/02564602.2015.1113146](https://doi.org/10.1080/02564602.2015.1113146)
- Rajkumar S, Goyal NK (2015c) Reliability analysis of multistage interconnection networks. *Qual Reliab Eng Int*. doi:[10.1002/qre.1941](https://doi.org/10.1002/qre.1941)
- Rajkumar S, Goyal NK (2015d) Reliable multistage interconnection network design. *Peer-to-Peer Netw Appl*. doi:[10.1007/s12083-015-0368-5](https://doi.org/10.1007/s12083-015-0368-5)
- Raponi PG, Andriolli N, Cerutti I, Torres D, Liboiron-Ladouceur O, Castoldi P (2013) Heterogeneous optical space switches for scalable and energy-efficient data centers. *J Lightwave Technol* 31(11):1713–1719
- Sengupta J, Bansal P (2001) High speed dynamic fault-tolerance'. *Procs IEEE Reg 10 Int Conf Electr Electron Technol* 2:669–675
- Sengupta J, Bansal P (2004) 'Performance analysis of static and dynamic fault-tolerant irregular networks'. In: *Proceedings of IEEE Region 10 International Conference on Electrical and Electronic Technology*, pp 5–8
- Shen X, Yang F, Pan Y (2001) Equivalent permutation capabilities between time-division optical omega networks and non-optical extra-stage omega networks. *IEEE/ACM Trans Netw (TON)* 9(4):518–524
- Shooman ML (2001) *Reliability of computer systems and networks: fault tolerance, analysis, and design*. Wiley Interscience, New York
- Verlinden S, Deconinck G, Coupe B (2012) Hybrid reliability model for nuclear reactor safety system. *Reliab Eng Syst Saf* 101:35–47
- Wang W, Mingxiao J (2004) 'Generalized decomposition method for complex systems'. In: *Reliability and maintainability annual symposium—RAMS*, pp 12–17
- Yang Y, Wang J (2004) A class of multistage conference switching networks for group communication. *IEEE Trans Parallel Distrib Syst* 15(3):228–243
- Yang Y, Wang J (2005) Routing permutations on baseline networks with node-disjoint paths. *IEEE Trans Parallel Distrib Syst* 16(8):737–746